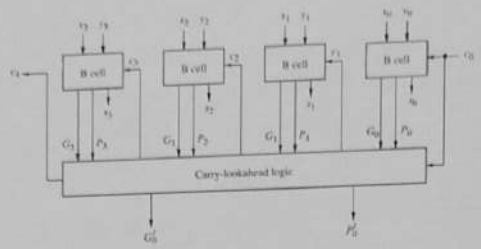
NATIONAL INSTITUTE OF TECHNOLOGY PATNA
Department of Computer Science and Engineering
END SEMESTER EXAMINATION, Jan-June, 2025
B. Tech (Computer Science and Engineering) 2nd Semester (Sections 1, 2, and 3)
***Course: Computer Organization* (CS24108)**
Full marks:60 Time: 3 hours

Answer *any six* questions

Q.no.	Question	Marks	CO	BL
1	a) Show the content of all the *cycles* and eventually the final product to multiply 1101 (multiplicand) with 1011 (multiplier) using sequential circuit binary multiplier.	05	CO2	Application
	b) How many logic gates are needed to build the 4-bit carry-lookahead adder shown in the following figure? 	05	CO2	Analysis and Application
2	a) Registers R4 and R5 contain the decimal numbers 2000 and 3000 before each of the following addressing modes is used to access a memory operand. What is the effective address (EA) in each case? i. 12(R4), ii. 12(R5), iii. 28(R4), iv. 28(R5)	05	CO3	Analysis and Application
	b) Consider a 16-bit processor in which the following appears in main memory, starting at location 200: 200 | Load to AC | Mode 201 | 500 202 | Next instruction The first part of the first word indicates that this instruction loads a value into an accumulator. The Mode field specifies an addressing mode and, if appropriate, indicates a source register; assume that when used, the source register is R1, which has a value of 400. There is also a base register that contains the value 100. The value of 500 in location 201 may be part of the address calculation. Assume that location 399 **contains the value** 999, location 400 includes the value 1000, and so on. Determine the effective address and the operand to be loaded for the following address modes: a. Direct, b. Immediate, c. Indirect, d. Register, e. Displacement or Index	05	CO3	Analysis
3	a) Write a RISC-style program to add n numbers using a subroutine, where parameters will be passed through the registers.	05	CO3	Application
	b) Draw the diagram of the datapath containing the control signals and explain the functionality of each control signal.	05	CO3	Remembering

4	a) A computer system uses 32-bit memory addresses and it has a main memory consisting of 1G bytes. It has a 4K-byte cache organized in the block-set-associative manner, with 4 blocks per set and 64 bytes per block. (a) Calculate the number of bits in each of the Tag, Set, and Word fields of the memory address. (b) Assume that the cache is initially empty. Suppose that the processor fetches 1088 words of four bytes each from successive word locations starting at location 0. It then repeats this fetch sequence nine more times. If the cache is 10 times faster than the memory, estimate the improvement factor resulting from the use of the cache. Assume that the LRU algorithm is used for block replacement.	06	CO4	Application
	b) Make a comparative analysis of merits and demerits of different cache mapping techniques.	04	CO4	Analysis
5	Consider a pipeline processor, P1, operating at 2 GHz with 4 stages: Instruction Fetch (IF), Instruction Decode (ID), Execute (EX), and Store. Each stage, except for the EX-stage, takes 2 cycles to complete its task. The EX-stage takes 4 cycles for multiplication (MUL) and division (DIV) operations but takes 3 cycles for any other operation. Now, let's compute the time taken by P1 to finish 5 instructions: ADD, MUL, DIV, Branch, and another DIV. Note that the MUL instruction has a data dependency on the ADD instruction, and the Branch instruction incurs a control hazard that results in 2 stall cycles. Additionally, assume that a new version of the P1 processor, labeled P2, has been developed. P2 features an inbuilt operand forwarding unit and a branch predictor that can completely eliminate all stall cycles. Now, let's calculate the speedup achieved by P2 over P1.	10	CO5	Analysis and Application
6	a) A processor can execute 1500 instructions in 30 microseconds. If these are three types of instruction (i) Arithmetic and Logical (70%), (ii) Branch and Control (15%), and Load and Store (15%). If each Arithmetic and Logical instruction needs 3 clock cycles, Branch and Control requires 4 clock cycles, and Load and Store requires 2 clock cycles to complete the task, then the CPU clock rate (in GHz) is	05	CO4	Analysis and Application
	b) Consider a computer system with Direct Memory Access (DMA) support. The DMA module can transfer two 8-bit characters in one CPU cycle from a device to memory using cycle stealing at regular intervals. With a processor operating at 2 MHz, if 0.5% of the processor's cycles are dedicated to DMA, the data transfer rate (in bits per second) for the device will be	05	CO4	Analysis and Application
7	a) Design a cache read operation. How does the *locality of reference* principle relate to using multiple memory levels?	05	CO4	Remembering
	b) Compare and contrast the instruction set complexity of RISC and CISC. How do these differences affect execution speed and memory usage?	05	CO3	Analysis
8	a) Discuss the techniques for resolving pipeline hazards (such as data, control, and structural hazards). How do these techniques impact the efficiency of pipelined processors?	5	CO5	Remembering
	b) Assume that 20 percent of the dynamic count of the instructions executed for a program are branch instructions. There are no pipeline stalls due to data dependencies. Static branch prediction is used with a not-taken assumption. i. Determine the execution times for two cases: when 30 percent of the branches are taken, and when 70 percent of the branches are taken. ii. Determine the speedup for one case relative to the other. Express the speedup as a percentage relative to 1.	5	CO5	Application